GLUEING DATA NEAR A POINT OF THE LIMIT (RETROCOMPACT VARIANT)

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Glueing of schemes. Schemes, glueing lemmas

Source reference. Original reference 01LG: sources/stacks-chow/tags.txt, source lines 2085–2085. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $U \subset S$ be a retrocompact open. Let $s \in S$ be a point in the complement of U. With $V = \mathrm{Spec}(\mathcal{O}_{S,s}) \cap U$ there is an equivalence of categories*

$$\mathrm{colim}_{s \in U' \supset U \text{ open}} \left\{ \begin{array}{c} X \\ \downarrow \\ U' \end{array} \right\} \longrightarrow \left\{ \begin{array}{ccccc} X' & \longleftarrow & Y' & \longrightarrow & Y \\ \downarrow & & \downarrow & & \downarrow \\ U & \longleftarrow & V & \longrightarrow & \mathrm{Spec}(\mathcal{O}_{S,s}) \end{array} \right\}$$

where on the left hand side the vertical arrow is of finite presentation and on the right hand side we consider commutative diagrams whose squares are cartesian and whose vertical arrows are of finite presentation.

Proof. Let $W \subset S$ be an open neighbourhood of s. By glueing of relative schemes, see Constructions, Section 01LG, the functor

$$\big\{ X \to U' = U \cup W \text{ of finite presentation} \big\} \longrightarrow \left\{ \begin{array}{ccccc} X' & \longleftarrow & Y' & \longrightarrow & Y \\ \downarrow & & \downarrow & & \downarrow \\ U & \longleftarrow & W \cap U & \longrightarrow & W \end{array} \right\}$$

is an equivalence of categories. We have $\mathcal{O}_{S,s} = \mathrm{colim}\, \mathcal{O}_W(W)$ where W runs over the affine open neighbourhoods of s. Hence $\mathrm{Spec}(\mathcal{O}_{S,s}) = \lim W$ where W runs over the affine open neighbourhoods of s. Thus the category of schemes of finite presentation over $\mathrm{Spec}(\mathcal{O}_{S,s})$ is the limit of the category of schemes of finite presentation over W where W runs over the affine open neighbourhoods of s, see Lemma 01ZM. For every affine open $s \in W$ we see that $U \cap W$ is quasi-compact as $U \to S$ is quasi-compact. Hence $V = \lim W \cap U$ is a limit of quasi-compact and quasi-separated schemes (see Lemma 01YX). Thus also the category of schemes of finite

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presentation over V is the limit of the categories of schemes of finite presentation over $W \cap U$ where W runs over the affine open neighbourhoods of s. The lemma follows formally from a combination of these results. $\square$